

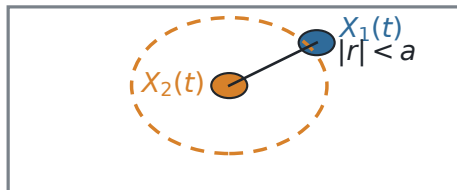
Spatial encounter channels and their modality boundary

(a) Finite-radius encounter

$$r = X_1 - X_2$$

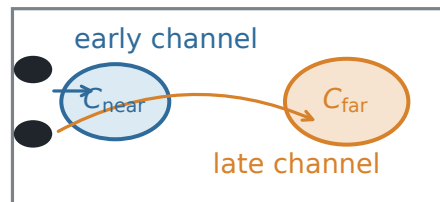
$$R = (D_2 X_1 + D_1 X_2) / (D_1 + D_2)$$

$$C(\eta) = \eta X_1 + (1 - \eta) X_2; \text{ bounded 2D: } \eta = 1/2$$



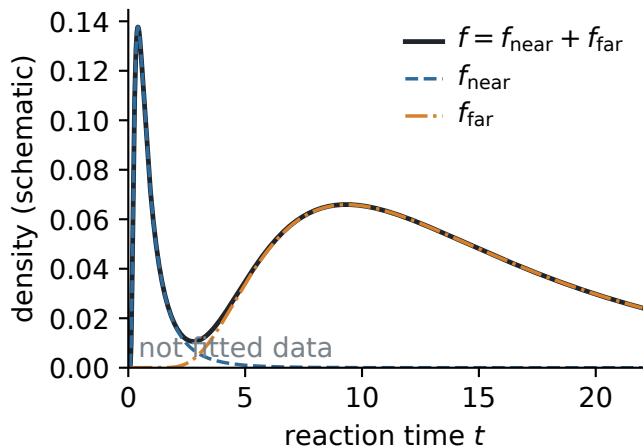
Reflecting domain; contact + catalyst required

(b) Pattern selects streams



$$K_a(\eta) = \sum_j \kappa_j \mathbb{1}[C_j](C(\eta)) \mathbb{1}[|r| < a]$$

(c) Channel mixture



(d) Generic modality fold

